



**WORKFORCEX-AI DRIVEN WORKFORCE DIGITAL TWIN TO OPTIMIZE
SKILL DEVELOPMENT**

A PROJECT REPORT

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ABSTRACT

The proposed **Smart Workforce Management System** is designed to provide an intelligent, centralized, and real-time approach to managing employees and organizational resources. In many workplaces, conventional attendance and workforce monitoring methods focus mainly on recording employee presence, while providing limited information for understanding workforce availability, employee capabilities, and task requirements. This creates difficulties in efficiently utilizing available personnel and making timely operational decisions. The proposed system addresses these challenges by integrating attendance monitoring with workforce profiling, real-time availability tracking, and skill-based task allocation.

The system automatically records employee attendance and maintains an updated representation of the workforce available within an organization. Each employee can be associated with relevant information such as role, skills, experience, department, and availability. This information is organized and presented through a centralized management platform, allowing authorized personnel to obtain a clear overview of the current workforce without depending on manual records. The system is intended to provide continuous visibility of employee status, thereby reducing administrative effort and improving the accuracy and reliability of workforce information.

A key feature of the proposed system is its **intelligent skill-matching and task-compatibility mechanism**. When a task or requirement is introduced, the system can compare the required skills and qualifications with the available employee profiles and identify personnel who are most suitable for the assigned responsibility. This helps organizations utilize their workforce more effectively, reduce unnecessary delays in task allocation, and ensure that employees are assigned according to their capabilities rather than availability alone. The system can also provide meaningful workforce insights through a dashboard, enabling supervisors and administrators to monitor attendance, availability, workforce distribution, and task-related information from a single platform.

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ABBREVIATIONS

Abbreviation	Full form
AI	Artificial Intelligence
ESP32	Espressif Systems 32-bit Microcontroller
IoT	Internet of Things
LED	Light Emitting Diode
GPIO	General Purpose Input/Output
RFID	Radio Frequency Identification
SPI	Serial Peripheral Interface
TFT	Thin-Film Transistor
UART	Universal Asynchronous Receiver/Transmitter
UID	Unique Identification Number
Wi-Fi	Wireless Fidelity
USB	Universal Serial Bus

CHAPTER 1

INTRODUCTION

In today's rapidly evolving organizational environment, effective workforce management has become an essential factor in improving productivity, operational efficiency, and resource utilization. Organizations often manage a large number of employees with different roles, skills, experience levels, and responsibilities. As the size and complexity of a workforce increase, maintaining accurate information about employee attendance, availability, capabilities, and task allocation becomes increasingly difficult. Conventional workforce management approaches generally depend on manual records or separate systems, which can result in delays, inconsistencies, and difficulty in obtaining real-time workforce information.

Attendance management is one of the fundamental aspects of workforce administration. Knowing whether an employee is present or available provides important information for day-to-day operations. However, attendance data alone does not provide sufficient insight into how effectively the available workforce can be utilized. An organization may have several employees present at a particular time, but assigning the right person to the right task requires additional information regarding their skills, experience, role, and suitability. Therefore, there is a need for a system that can go beyond basic attendance tracking and provide meaningful workforce intelligence.

The proposed **Smart Workforce Management System** is developed to address these challenges through an integrated and centralized approach. The system enables automated employee identification and attendance recording while maintaining an up-to-date view of workforce availability. Employee information can be organized according to relevant attributes such as department, designation, skills, experience, and other professional characteristics. This information allows administrators or supervisors to obtain a clearer understanding of the workforce currently available within the organization.

One of the significant aspects of the proposed system is its ability to support **skill-based workforce allocation**. Different tasks may require different combinations of technical skills, experience, or qualifications. The system can compare the requirements of a particular task with the available employee profiles and determine suitable candidates based on their compatibility. This approach helps reduce the dependence on manual decision-making and enables organizations to make faster and more informed workforce allocation decisions.

The system also provides a **centralized dashboard** through which workforce information can be monitored and analyzed. Important information such as attendance status, employee availability, workforce distribution, task requirements, and skill compatibility can be presented in an organized and easily understandable manner. This provides management with a consolidated view of workforce conditions and supports efficient monitoring and planning.

The proposed approach therefore shifts workforce management from a conventional **attendance-oriented model to a data-driven workforce intelligence model**. Instead of treating attendance as an isolated record, the system utilizes it as one of the key inputs for understanding workforce availability and supporting operational decisions. By combining real-time monitoring, employee profiling, skill analysis, and task compatibility, the system aims to improve workforce utilization and organizational coordination.

The solution is designed to be **efficient, scalable, and adaptable** to different organizational environments. It can be applied in offices, industries, educational institutions, service organizations, and other workplaces where employee availability and skill-based resource allocation play an important role. Ultimately, the proposed system aims to provide a smarter and more systematic method of managing human resources while reducing manual effort, improving accuracy, and supporting better decision-making.

1.1 Background of Study

1.1.1 Traditional Workforce Management

Traditional workforce management primarily involves maintaining employee attendance, work schedules, departmental records, and task assignments. In many organizations, these activities are handled through manual registers, spreadsheets, or independent software applications. Although such methods can support basic administration, they become difficult to manage as the number of employees and organizational activities increase.

1.1.2 Need for Real-Time Workforce Monitoring

The availability of employees can change continuously due to attendance, leave, shifts, or task assignments. Therefore, relying on static or manually updated records may not provide an accurate representation of the workforce at a given moment. Real-time workforce

monitoring allows administrators to obtain updated information about employee availability and make timely operational decisions.

1.1.3 Skill-Based Workforce Allocation

Employees possess different skills, qualifications, experience, and areas of expertise. Assigning a task solely based on employee availability may not always result in effective resource utilization. Skill-based allocation provides a systematic approach to identifying employees whose capabilities are relevant to the requirements of a particular task.

1.1.4 Workforce Intelligence and Decision Support

Modern workforce management can utilize employee information to generate meaningful insights rather than simply storing attendance records. By connecting attendance, availability, employee skills, and task requirements, organizations can obtain a better understanding of their workforce. Such information can support supervisors in making faster and more informed decisions.

1.1.5 Centralized Workforce Management

A centralized management platform provides a single location for accessing important workforce information. Instead of depending on multiple records or systems, administrators can monitor employee status, workforce availability, skills, and task-related information through an organized interface. This improves accessibility, coordination, and overall workforce visibility.

1.2 Statement of Problem

1.2.1 Limitations of Conventional Attendance Systems

Conventional attendance systems mainly concentrate on recording employee presence and absence. They provide limited information about employee skills, availability, and suitability for organizational tasks. As a result, attendance data is not fully utilized for workforce planning.

1.2.2 Manual Workforce Monitoring

Manual methods of maintaining employee records require continuous administrative effort and may result in outdated or inaccurate information. Checking employee availability manually can also become time-consuming when dealing with a large workforce.

1.2.3 Difficulty in Selecting Suitable Employees

Identifying the right employee for a particular task can be challenging when employee skills and task requirements are maintained separately. Manual selection may depend heavily on individual judgment and may not consistently consider all relevant employee capabilities.

1.2.4 Inefficient Utilization of Skills

When task allocation is based primarily on availability rather than capability, employees may be assigned to tasks that do not match their expertise. This can result in inefficient utilization of human resources and may affect productivity.

1.2.5 Lack of Centralized Workforce Information

The absence of a unified platform for attendance, employee information, skills, availability, and task requirements makes it difficult for management to obtain a complete view of the workforce. A centralized solution is therefore required to improve workforce monitoring and decision-making.

1.3 Objective of Study

The primary objective of this study is to develop a **Smart Workforce Management System** that enables organizations to monitor, analyze, and utilize their workforce more effectively. The system aims to automate the process of employee attendance monitoring and provide real-time information regarding workforce availability. It is intended to maintain organized employee profiles containing relevant information about roles, skills, experience, and other attributes that can contribute to effective workforce planning. Another major objective is to implement skill-based matching between employee capabilities and task requirements so that suitable personnel can be identified for specific responsibilities. The system also aims to support compatibility analysis and improve task allocation by considering both employee capabilities and availability. In addition, the study focuses on providing a centralized dashboard through which administrators and supervisors can access important workforce information in an organized manner.

CHAPTER 2

LITERATURE SURVEY

RFID Technology Based Attendance Management System (S. Nainan, R. Parekh, and T. Shah, 2013)

The authors proposed an RFID-based attendance management approach for automating the identification and attendance-recording process. The study explains the use of RFID technology for automatic identification and highlights its advantages such as scalability, security, and reduced manual effort. The work demonstrates that RFID can provide a faster and more reliable alternative to conventional attendance methods, forming a foundation for automated employee attendance monitoring.

IoT Based Smart Attendance System Using RFID: A Systematic Literature Review (K. Ishaq and S.Bibi,2023)

The authors conducted a systematic review of 21 research studies related to IoT-based attendance systems using RFID technology. The study highlights how RFID combined with IoT can automate attendance recording, reduce time consumption, prevent common problems associated with manual attendance, and enable real-time attendance monitoring. The review also identifies challenges including security vulnerabilities, technical issues, and implementation costs, providing useful considerations for developing reliable IoT-based attendance systems.

Workforce Planning Incorporating Skills: State of the Art (P. De Bruecker, J. Van den Bergh, J. Beliën, and E. Demeulemeester, 2015)

The authors presented a comprehensive review of workforce planning techniques that incorporate employee skills. The study emphasizes that workforce planning should consider the different capabilities of employees instead of treating all workers as equivalent resources. It discusses factors such as skill requirements, skill substitution, cross-training, efficiency, and workload. This work provides an important basis for developing a system that matches employees with tasks according to their skills and availability.

Employee-Task Assignments for Organization Modeling: A Review of Models and Applications (C. Kandemir and H. A. H. Handley, 2014)

The authors reviewed computational approaches for assigning organizational tasks to employees based on their qualifications and capabilities. The study explains that different

tasks require different expertise and that assigning tasks to the most qualified available personnel can improve organizational performance. However, it also highlights the importance of workload distribution because continuously assigning tasks to the same highly qualified employees can cause overloading. These findings are relevant to developing a workforce system that considers both employee suitability and availability while recommending personnel for specific tasks.

Optimal Complex Task Assignment in Service Crowdsourcing (F. Tang, 2020)

The author proposed a multi-skill-based task assignment method for matching complex tasks with suitable workers. The study introduces a weighted multi-skill tree to represent relationships between different skills and develops an assignment algorithm that considers both worker suitability and temporary unavailability. Experimental results demonstrate improved task-assignment performance compared with existing approaches. This research supports the use of skill-based compatibility calculations for selecting suitable employees for particular tasks in the proposed workforce management system.

IoT-Based Employee Attendance and Monitoring Systems (Recent IoT-Based Approaches, 2023–2026)

Recent research has increasingly combined RFID readers, microcontrollers, wireless communication, databases, and web-based interfaces to create connected attendance systems. These approaches demonstrate that employee identification can be performed locally using RFID while attendance information can be transmitted to a centralized platform for monitoring and record management. Such systems provide the technological foundation for integrating an RFID reader, embedded controllers, wireless connectivity, and a real-time dashboard in an employee-management application.

Skill-Based Workforce Assignment and Scheduling (Recent Research)

Recent workforce scheduling studies have further emphasized the importance of matching employee competencies with task requirements while considering employee availability and workload. These approaches demonstrate that workforce efficiency can be improved when task allocation is based on measurable compatibility rather than simple availability. However, many existing approaches rely on mathematical optimization models and are designed for large-scale organizational environments, making them relatively complex for direct implementation in a low-cost embedded prototype. This creates an opportunity for a simpler compatibility-based approach suitable for real-time workforce management.

Research Gap

From the above studies, it can be observed that **RFID and IoT-based research mainly concentrates on attendance identification and monitoring**, while **workforce-planning research focuses primarily on skill-based task allocation and scheduling**. These functions are generally studied as separate problems. Therefore, there is a need for an integrated system that combines **RFID-based employee identification, real-time attendance monitoring, employee skill information, task requirements, compatibility-based employee selection, and centralized dashboard visualization**. The proposed system addresses this gap by combining embedded hardware based on **STM32 and ESP32** with RFID identification and a software dashboard for intelligent workforce management.

CHAPTER 3

EXISTING SYSTEM

3.1 Introduction

Workforce management has evolved from conventional attendance recording into comprehensive digital platforms that support employee time tracking, scheduling, absence management, workforce planning, reporting, and labor optimization. Modern workforce-management solutions are increasingly designed to provide organizations with centralized information for managing employees and operational requirements. Industry research identifies time and attendance, scheduling, resource allocation, task management, analytics, and workforce optimization as major capabilities of contemporary workforce-management platforms. [ISG Research](#)

Traditional approaches began with manual registers, paper-based time cards, and mechanical time-recording systems. As organizations expanded, these methods became increasingly difficult to maintain because they required manual collection, verification, and processing of employee records. Recent research continues to identify issues such as incomplete records, fragmented information, manual exception handling, and limited real-time visibility as important challenges in workforce management. [sciencepg.com](#)

Current digital systems have addressed many of these limitations by automating attendance and integrating workforce information with scheduling, payroll, leave management, and reporting. For example, commercial workforce-management platforms provide capabilities for time capture, attendance monitoring, employee availability, scheduling, absence management, and workforce analytics. [ADP](#) However, the requirements of modern organizations increasingly extend beyond determining whether an employee is available. Effective workforce utilization also requires identifying **which available employee is most suitable for a particular task based on skills, experience, workload, and task requirements**. Research on employee-task allocation demonstrates that assigning tasks appropriately is itself a complex optimization problem, where unsuitable assignments can negatively affect efficiency and workforce utilization. [ScienceDirect](#)

Therefore, although existing workforce-management systems provide strong support for attendance, scheduling, and workforce administration, there remains scope for developing focused solutions that connect **real-time workforce availability with employee capabilities**

and task compatibility in a simple and accessible manner. This forms the basis for the proposed system.

3.2 Existing Workforce Management Systems

Existing workforce-management systems can broadly be categorized into conventional attendance systems, digital attendance and HR platforms, and comprehensive workforce-management suites.

Conventional attendance systems primarily focus on recording employee entry and exit information. Earlier approaches used paper registers, punch cards, or similar time-recording methods. Such systems are relatively simple but require considerable manual effort for maintaining records and generating reports. They also provide little information beyond an employee's attendance status. Research on time-and-attendance systems identifies manual correction, incomplete records, data inconsistency, and lack of real-time visibility among the limitations associated with legacy processes. sciencepg.com

Digital attendance and HR management systems have significantly improved this process by automating attendance capture and connecting it with employee records, leave management, payroll, and reporting. These systems reduce paperwork and improve the speed and consistency of administrative operations. Modern workforce-management software can collect and analyze time and attendance information, manage absences, create schedules, and provide workforce-related analytics. [ADP](#)

Enterprise workforce-management suites provide broader functionality by integrating attendance, scheduling, labor optimization, task management, forecasting, analytics, and employee management. Current industry research describes these platforms as evolving beyond basic timekeeping toward advanced capabilities such as skills-based optimization, task and activity management, real-time analysis, and AI-assisted workforce planning. [ISG Research](#)

Although these platforms provide extensive functionality, their primary focus is generally on comprehensive enterprise workforce administration, scheduling, compliance, payroll integration, and operational optimization. For a project focused specifically on **real-time**

attendance combined with employee skill matching and task compatibility, there is an opportunity to create a more focused and easily understandable solution.

3.3 Limitations of the Existing System

The major limitation identified from the existing systems is not the absence of automation, but the **fragmentation of workforce information and the difference between monitoring employees and intelligently utilizing them**. Many attendance-oriented systems are primarily concerned with recording presence, while task allocation may remain a separate managerial activity. A recent survey of attendance and payroll systems similarly identifies that many existing solutions concentrate on attendance or payroll rather than providing an integrated workforce-management approach. [IJERT](#)

Another limitation is that employee availability does not necessarily indicate employee suitability. An employee may be present and available but may not possess the skills required for a particular task. Effective task allocation therefore requires consideration of employee capabilities in addition to attendance status. Research on employee positioning and workload allocation demonstrates that employee-task assignment involves multiple factors and that inappropriate assignments can reduce efficiency. [ScienceDirect](#)

Existing enterprise platforms can address some of these requirements through advanced optimization and scheduling capabilities, but such systems are often designed as broad HR and workforce-management ecosystems. Their complexity, scale, and wide range of features may not be necessary for smaller organizations or focused applications where the primary requirement is to obtain a clear real-time view of available employees and identify suitable personnel for a particular task. Current workforce-management research also emphasizes that advanced optimization depends on effective integration of workforce data and consistent system usage. [ISG Research](#)

Hence, the key gap addressed by the proposed project is the development of a **focused workforce-management platform that connects attendance and real-time availability with employee skills and task requirements**, presenting the resulting information through a centralized and easy-to-understand interface.

CHAPTER 4

PROPOSED SYSTEM

4.1 Introduction

The proposed **Smart Workforce Management System** is designed to provide a unified and intelligent approach to managing employee attendance, workforce availability, employee capabilities, and task allocation. Unlike conventional systems that primarily concentrate on recording attendance or maintaining employee information, the proposed system uses workforce data as the basis for operational decision-making. It provides a centralized platform through which the current workforce can be monitored, employee profiles can be maintained, and suitable personnel can be identified according to the requirements of a particular task. The system is intended to provide organizations with a clear and continuously updated view of their available workforce while reducing the effort involved in manually collecting and analyzing employee information.

Fig 4.1 Hardware image of AI-based public crowd management system

4.2 System Overview

The proposed system operates as an integrated workforce-management platform in which employee identification and attendance information are automatically captured and updated within the system. Once an employee is identified, the relevant attendance status is recorded and associated with the employee's profile. The system maintains information such as employee identity, department, designation, skills, experience, and availability, allowing the workforce to be viewed as a structured and searchable resource rather than simply a collection of attendance records.

The collected information is made available through a centralized dashboard that provides management with real-time workforce visibility. Administrators or authorized users can monitor employee presence, availability, workforce distribution, and other relevant information from a single interface. This eliminates the need to depend on multiple records or manually consolidate information before making workforce-related decisions.

4.3 Employee Management

The system maintains a structured profile for each employee containing relevant professional information. Employee profiles can include their role, department, technical and non-technical skills, experience, qualifications, and other attributes required for workforce planning. These profiles form the foundation for the system's skill-based analysis.

Maintaining this information in a centralized manner allows employee capabilities to be considered whenever a new task or workforce requirement is introduced. It also makes it easier for administrators to identify the available skills within the organization and understand how the workforce can be utilized effectively.

4.4 Automated Attendance and Availability Management

A major function of the proposed system is automated attendance management. Employee attendance is recorded when the employee is identified, reducing the need for manual entry and minimizing errors associated with conventional attendance recording. The system maintains the attendance status of employees and updates their availability accordingly.

Rather than treating attendance as an isolated record, the proposed system uses attendance information as an input for workforce planning. This enables management to distinguish between employees who are currently available and those who are absent or otherwise unavailable, providing a more accurate representation of the workforce at a given time.

4.5 Skill Matching and Task Allocation

One of the primary features that differentiates the proposed system is its **skill-based task matching capability**. When a task is created, the system considers the skills and competencies required to complete that task. These requirements are compared with the skill profiles of employees who are currently available.

The system evaluates the degree to which an employee's capabilities correspond to the requirements of the task. Based on this comparison, suitable employees can be identified and prioritized. This approach assists supervisors in selecting personnel based not only on availability but also on their ability to perform the required work.

The matching process can consider multiple factors, such as required skills, employee expertise, experience, role, and current availability. This provides a more systematic

approach to task allocation and can help improve the utilization of specialized skills within an organization.

4.6 Compatibility Analysis

The proposed system can provide a **compatibility assessment** between an employee and a task. Instead of presenting only a list of available employees, the system can determine how closely each employee matches the requirements of the task.

For example, if a task requires several specific skills, employees possessing a greater number of the required skills can receive a higher compatibility score. The results can then be presented in a ranked manner, allowing management to quickly identify the most suitable candidates. This reduces the time required to manually compare employee profiles and task requirements.

4.7 Centralized Dashboard

The proposed system includes a centralized dashboard that acts as the primary interface for workforce monitoring and management. It provides management with organized information regarding employee attendance, current availability, workforce distribution, employee skills, task requirements, and matching results.

The dashboard is intended to convert workforce data into easily understandable information. Instead of requiring administrators to examine individual records, important information can be viewed through appropriate summaries, tables, indicators, and visual representations. This supports faster monitoring and improves the overall visibility of workforce conditions.

4.8 Real-Time Workforce Monitoring

The system is designed to maintain updated workforce information so that management can obtain a current view of employee availability. Changes in attendance status can be reflected in the workforce information, allowing task allocation decisions to be made using more current data.

Real-time monitoring is particularly useful in environments where employee availability changes frequently and tasks need to be assigned within a limited time. It enables supervisors to respond to workforce changes without repeatedly checking or updating separate records.

4.9 Flowchart

Here, we have attached the flowchart of the project. The flowchart is a visual representation that makes it easier to analyze.

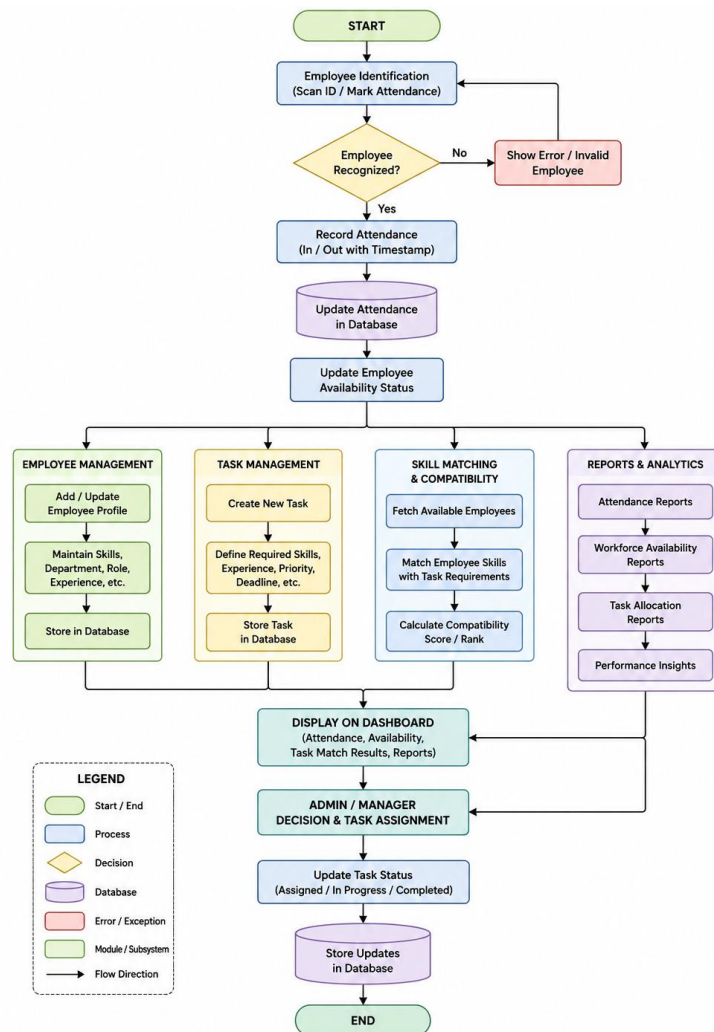


Fig 4.1 Flow Chart of AI driven workforce digital twin for skill development

4.4 Flowchart Process

Employee Identification

The process begins with employee identification. When an employee interacts with the system, the provided identification information is captured and compared with the registered employee records. This enables the system to determine whether the employee is authorized and already registered within the workforce database.

Employee Verification

After identification, the system verifies the employee details. If the employee is not recognized, the system displays an appropriate error or invalid-employee indication and redirects the process back to the identification stage. If the employee is successfully verified, the system proceeds with attendance processing.

Attendance Recording

Once the employee is verified, the system records the attendance information along with the relevant date and time. This eliminates the need for manual attendance entry and ensures that employee attendance records are maintained systematically.

Database Update

The recorded attendance information is stored in the central database. The system also updates the employee's current availability status based on the recorded attendance. This ensures that workforce information remains synchronized with the latest attendance activity.

Employee Profile Management

The system maintains a structured profile for each employee. Information such as employee details, department, designation, skills, qualifications, and experience can be stored and updated. These profiles provide the information required for workforce analysis and skill-based task allocation.

Task Management

Administrators can create and manage organizational tasks through the system. Each task can be associated with specific requirements such as required skills, experience, priority, and completion deadline. The task information is stored for further processing and allocation.

Skill Matching

When a task is created, its requirements are compared with the skills available within the employee profiles. The system considers employees who are currently available and evaluates their capabilities against the requirements of the task.

Compatibility Analysis

The system performs compatibility analysis to determine how closely each available employee matches the requirements of the selected task. Based on the matching results, employees can be assigned a compatibility score or ranking. This allows the most suitable candidates to be identified more efficiently.

Workforce Reports and Analytics

The system generates workforce-related information such as attendance records, employee availability, task allocation information, and other relevant insights. These reports assist administrators and managers in understanding workforce conditions and making informed decisions.

Dashboard Visualization

The processed information is presented through a centralized dashboard. The dashboard provides an organized view of attendance, employee availability, employee information, task requirements, skill-matching results, and reports. This allows authorized users to monitor workforce conditions from a single platform.

Task Allocation

Based on the employee availability and compatibility results, the administrator or manager can select a suitable employee for the task. The system therefore supports task allocation based on both **availability and capability**, rather than relying solely on manual selection.

Task Status Update

After allocation, the status of the task can be updated according to its progress. The task may move through stages such as **Assigned, In Progress, and Completed**. These updates provide continuous visibility into the progress of organizational activities.

Data Storage and Completion

All relevant attendance, employee, task, matching, and status information is maintained in the database for future reference and analysis. After the required updates are stored, the current workflow reaches its completion state. The system can subsequently begin a new identification or task-management cycle, allowing the process to operate continuously.

CHAPTER 5

HARDWARE DESCRIPTION AND SOFTWARE DESCRIPTION

5.1 Hardware Requirements

5.1.1 ESP32

The **ESP32** is a powerful and low-cost microcontroller with built-in **Wi-Fi and Bluetooth** connectivity. It is widely used in IoT applications for wireless data communication and remote monitoring. Its compact design, multiple communication interfaces, and low power consumption make it suitable for connected embedded systems. In this project, it enables wireless communication between the embedded system and the workforce management dashboard.



Fig 5.1 ESP32

The **ESP32** is a low-cost, highly integrated microcontroller and wireless communication platform developed by Espressif Systems, featuring a powerful processing architecture along with built-in **Wi-Fi and Bluetooth connectivity**. It provides multiple digital and analog interfaces, communication peripherals such as UART, SPI, and I²C, and sufficient processing capability for IoT-based applications. In the proposed **Smart Workforce Management System**, the ESP32 is primarily used as the **wireless communication unit** that connects the embedded system with the centralized workforce management dashboard. After the employee-related information is processed, the ESP32 enables the system to transmit attendance and workforce data over a wireless network to the dashboard, allowing administrators to access updated information without requiring a wired connection. Its built-in Wi-Fi capability eliminates the need for an additional external wireless communication module, thereby simplifying the overall system architecture and reducing implementation

complexity. The ESP32 is particularly suitable for this project because the system requires real-time data transmission, remote monitoring, and communication between the physical workforce identification unit and the digital dashboard.

5.1.3 RFID MODULE

The **RFID (Radio Frequency Identification) module** is an electronic device used to wirelessly read identification data from RFID cards or tags using radio-frequency signals. It enables contactless identification without requiring physical insertion or direct contact with the reader. RFID modules are widely used in attendance systems, access control, inventory tracking, and identification applications.



Fig 5.2 RFID MODULE

In the proposed **Smart Workforce Management System**, the RFID module is used to provide **fast, contactless, and automated employee identification**. When an employee presents their registered RFID card near the reader, the module detects the card and retrieves its unique identification number. This information is then processed by the system to identify the corresponding employee and record their attendance. The use of RFID eliminates the need for manual attendance entry and reduces the possibility of errors associated with handwritten or manually entered records. It also provides a quick identification process, which is important when multiple employees need to register their attendance within a short period. The RFID module therefore serves as the primary identification interface through which employee presence is captured and converted into digital workforce information for further processing and monitoring.

In this project, each employee is assigned a **unique RFID card** that acts as their digital identification credential. When the employee presents the card to the RFID reader, its unique identification number is captured and associated with the corresponding employee profile stored in the system. This allows attendance to be recorded automatically and helps maintain an accurate record of employee presence and availability. RFID cards are preferred because they are **simple to use, reusable, contactless, inexpensive, and easy to assign to individual employees**. They also eliminate the need for employees to manually enter identification details, making the attendance process faster and more convenient. The unique identification provided by each card also enables the system to link attendance information with the employee's profile, which can subsequently be used for workforce availability monitoring and other workforce-management functions.

5.1.4 TFT display

The **TFT (Thin-Film Transistor) display** is a type of color LCD display that provides clear and detailed visual output. It is capable of displaying text, numbers, graphics, icons, and other interface elements with good brightness and resolution. TFT displays are commonly used in embedded systems where a more informative and interactive visual interface is required.



Fig 5.3 TFT display

In the proposed **Smart Workforce Management System**, the TFT display is used as the **local visual interface** for providing immediate feedback to employees and system operators. After an employee is identified, the display can show relevant information such as employee identification, attendance confirmation, date, time, and current status. It can also provide system messages and notifications, allowing users to understand the result of an operation without depending entirely on the centralized dashboard. The TFT display is selected because

its **color output, larger display area, and ability to present both text and graphical information** make it suitable for creating a clear and user-friendly interface. It improves the usability of the system by providing instant on-site feedback during employee identification and attendance operations.

5.1.5 LED

An LED (Light Emitting Diode) is a semiconductor device that emits light when an electric current passes through it. It is widely used in electronic systems due to its low power consumption and high efficiency. LEDs are commonly used as indicators, displays, and lighting components in various applications.



Fig 5.4 LED

In the proposed **Smart Workforce Management System**, LEDs are used as **visual status indicators** to provide quick feedback during system operation. Different LED indications can represent conditions such as successful employee identification, attendance confirmation, invalid identification, or system activity. This allows users to understand the immediate status of an operation without having to continuously check the display. LEDs are used because they provide a **simple, fast, and easily recognizable indication**, improving the usability and responsiveness of the system.

5.1.6 Buzzer

The **buzzer** is an electronic sound-producing device used to generate audible signals or alerts in an electronic system. It is commonly used to indicate events, confirmations, warnings, or errors through different sound patterns.



Fig 5.5 BUZZER

In the proposed **Smart Workforce Management System**, the buzzer is used to provide **audible feedback during employee identification and attendance operations**. A short beep can indicate successful identification and attendance registration, while a different or repeated sound can indicate an invalid card or unsuccessful operation. This provides immediate feedback to the employee without requiring them to look at the display every time. The buzzer therefore improves the **user interaction, accessibility, and responsiveness** of the system and provides a simple indication of important system events

5.2 Software Requirements

5.2.1 Functional Requirements

The system must be capable of automatically identifying employees using the RFID cards provided to registered personnel. When an employee places the RFID card near the RFID reader, the system must read the unique identification number (UID) of the card and compare it with the registered employee information. The system must record the employee's attendance along with the corresponding date and time and update the attendance status in the system.

The system must provide real-time communication between the embedded hardware and the workforce management dashboard. The ESP32 must transmit the collected attendance and employee information through a wireless network to the backend or database. The system should provide immediate feedback to the employee through the TFT display, LED and buzzer to indicate successful identification, invalid cards, or other system conditions.

The system must be capable of maintaining employee information including employee ID, name, department, skills, experience and current availability status. The system should also maintain task-related information such as task ID, task description, required skills and task priority. Based on the available employee information and task requirements, the system must calculate the compatibility between employees and assigned tasks and identify the most suitable employee for a particular task.

The system must provide a dashboard for administrators or authorized users to monitor the workforce in real time. The dashboard should display employee attendance, current availability, employee details, task information, skill compatibility and recommended employee assignments. The system should also maintain historical attendance and task-assignment information for future reference and analysis.

5.2.2 Non-Functional Requirements

The system should provide reliable and accurate employee identification and attendance recording. The response time between RFID card detection and displaying the attendance status should be sufficiently low for practical use. The system should be easy to operate, maintain and expand with additional employees, RFID cards, skills and tasks.

The system should provide secure communication and data storage to prevent unauthorized access to employee and attendance information. The dashboard should have a simple and user-friendly interface so that administrators can easily monitor live workforce information and understand employee-task compatibility. The system should also be scalable so that additional sensors, employees, tasks and intelligent decision-making features can be incorporated in future versions.

5.2.3 Software Modules

RFID Identification Module

The RFID Identification Module is responsible for communicating with the RFID reader and obtaining the unique identification number of each employee RFID card. The received UID is compared with the registered employee database to authenticate the employee. Based on the verification result, the module generates the appropriate attendance response.

Attendance Management Module

The Attendance Management Module processes the authenticated RFID information and records employee attendance along with the date and time. It maintains the employee's present or absent status and transfers the updated information to the database and dashboard for real-time monitoring.

Embedded Control Module

The Embedded Control Module manages the operation of the STM32F401RE and connected peripherals. It handles RFID communication, display operations, LED and buzzer indications, and communication with the ESP32. The module ensures proper coordination between the different hardware components of the prototype.

Wireless Communication Module

The Wireless Communication Module uses the ESP32 to establish Wi-Fi connectivity and transfer information between the embedded system and the server or dashboard. It is responsible for sending attendance information and receiving relevant employee or task-related data.

Employee Management Module

The Employee Management Module stores and manages employee information such as employee ID, name, department, RFID UID, skills, experience and availability. It provides the required employee data for attendance monitoring and task allocation.

Skill Matching and Task Assignment Module

The Skill Matching and Task Assignment Module compares the skills required for a particular task with the skills possessed by available employees. A compatibility score can be calculated based on the matching skills and other parameters such as availability, experience and task requirements. The module then recommends the most suitable employee for the assigned task.

Dashboard and Monitoring Module

The Dashboard and Monitoring Module provides a graphical interface for administrators to view real-time workforce information. It displays live attendance, employee availability, task details, skill compatibility and recommended employee assignments. The module also provides access to historical records and statistical information for workforce analysis.

Database Management Module

The Database Management Module is responsible for storing and retrieving employee information, RFID records, attendance data, skills, tasks and assignment results. It ensures that the information generated by the system is maintained systematically and can be accessed by the dashboard whenever required.

Code

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <SPI.h>
#include <MFRC522.h>
#include <Adafruit_GFX.h>
#include <Adafruit_ST7735.h>
#include <ArduinoJson.h>

// =====
// WIFI
// =====

const char* WIFI_SSID = "ev....";
const char* WIFI_PASSWORD = "123....";

// =====
// WORKFORCEX BACKEND
// =====

const char* SERVER_IP = "10.89.92.65";

const char* ATTENDANCE_URL =
    "http://10.89.92.65:8000/attendance/rfid-scan";

// =====
// RC522
// =====

#define RFID_SS    21
```

```

#define RFID_RST 22

// =====
// TFT
// =====

#define TFT_CS 5
#define TFT_RST 4
#define TFT_DC 2

// =====
// STATUS LEDS + BUZZER
// =====

#define GREEN_LED 25
String aiReason;

// =====
// CURRENT TASK
// =====

String taskName;
String taskProject;
String taskStatus;
String taskPriority;
String taskDeadline;
String taskRequiredSkill;

int taskProgress = 0;

// =====
// SKILL DATA
// =====

String skillNames[10];
int skillLevels[10];
int skillCount = 0;

// =====

```

```

// TASK DATA
// =====

String taskNames[10];
String taskStatuses[10];
int taskProgresses[10];
String taskPriorities[10];
String taskDeadlines[10];

int taskCount = 0;

// =====
// PAGE CONTROL
// =====

int currentPage = 0;

unsigned long lastPageChange = 0;

const unsigned long PAGE_INTERVAL = 2500;

bool employeeLoaded = false;

// =====
// SPI CONTROL
// =====

void selectRFID()
{
    digitalWrite(TFT_CS, HIGH);
    digitalWrite(RFID_SS, LOW);
}

void deselectRFID()
{
    digitalWrite(RFID_SS, HIGH);
}

```



```

void selectTFT()
{
    digitalWrite(RFID_SS, HIGH);
    digitalWrite(TFT_CS, LOW);
} // =====

// PAGE 4 - SKILLS
// =====

void showSkillsPage()
{
    drawHeader("SKILLS");

    int y = 45;

    for (int i = 0; i < skillCount && i < 5; i++)
    {
        selectTFT();

        tft.setTextColor(ST77XX_WHITE);

        tft.setCursor(4, y);

        tft.print(skillNames[i]);

        tft.setTextColor(ST77XX_CYAN);

        tft.print(" ");

        tft.print(skillLevels[i]);

        tft.println("%");

        deselectTFT();

        y += 15;
    }
}

printLine(

```

```

    103,
    "OVERDUE: ",
    String(overdueTasks)
);

println(
    118,
    "PERFORMANCE: ",
    String(employeePerformance, 1) + "%"
);
}

// =====

```

5.2.4 Hardware and Software Integration

Hardware and Software Integration

The proposed workforce management system integrates embedded hardware components with software modules to provide automated employee identification, real-time attendance monitoring, task information display and intelligent workforce management. The **ESP32** acts as the **main and only embedded controller** of the system, providing control and communication between all hardware components and the software platform. It interfaces directly with the **RFID reader, TFT display, LED and buzzer**.

The RFID reader detects the employee's RFID card and sends its unique identification number to the ESP32. The ESP32 processes the RFID data and verifies the employee's identity with the information stored in the backend or database through Wi-Fi communication. After successful identification, the ESP32 provides immediate feedback using the **TFT display, LED and buzzer**. The TFT display can present relevant employee information such as employee name, assigned task, task completion percentage, task level and attendance status.

The ESP32 provides **wireless communication between the embedded hardware and the software platform** using its built-in Wi-Fi capability. Once an employee scans their RFID card, the ESP32 sends the identification and attendance information to the backend/database. The software system maintains employee details, RFID information, attendance records, skills, availability, assigned tasks and task requirements. The workforce management

dashboard retrieves this information and presents it to the administrator for real-time monitoring and management.

The system also integrates a **Skill Matching and Task Assignment Module**, which compares the skills, availability and capabilities of employees with the requirements of available tasks. Based on the matching criteria, the software calculates a compatibility score and recommends the most suitable employee for a particular task. Once a task is assigned, the relevant task information is made available to the ESP32, allowing the employee to view their assigned task and progress directly on the TFT display after RFID authentication.

The **ESP32 serves as the central embedded controller**, eliminating the need for a separate STM32 controller and simplifying the overall hardware architecture. This single-controller approach reduces hardware complexity, wiring requirements and communication overhead while allowing the system to perform RFID authentication, user feedback, wireless data transmission and display control. The integration therefore combines **physical employee identification, real-time attendance monitoring, task visualization and intelligent workforce management** into a single connected platform.

CHAPTER 6

RESULT AND DISCUSSIONS

The developed **WORKFORCEX** prototype successfully integrates RFID-based employee identification, embedded hardware, a FastAPI backend, database management, and a web-based digital twin dashboard into a unified workforce management system.

The RFID module successfully identifies registered employees through their unique RFID cards. Once an employee is authenticated, the ESP32 processes the RFID data and communicates with the backend system. The TFT display provides immediate feedback such as employee identity and access status, demonstrating the feasibility of real-time employee identification at the hardware level.

On the software side, the developed dashboard provides a centralized view of employee profiles, departments, roles, skills, project assignments, workload, and task information. The system also enables project and task creation and maintains the relationship between employees, skills, and project requirements.

The **AI-based employee allocation module** further extends the system by considering factors such as employee skills, current workload, task requirements, priority, and deadline urgency when generating allocation recommendations. This provides a structured approach to workforce deployment rather than relying only on manual assignment.

The integration of the hardware and software components demonstrates a complete data flow:

RFID Card → RC522 → ESP32 → FastAPI Backend → Database → Digital Twin Dashboard

During prototype testing, the system demonstrated successful RFID detection, employee verification, backend communication, and visualization of employee and project information. The prototype therefore validates the basic feasibility of using a **digital-twin-based approach for workforce visibility and skill deployment**.



Fig 6.1 hardware (physical identification layer)

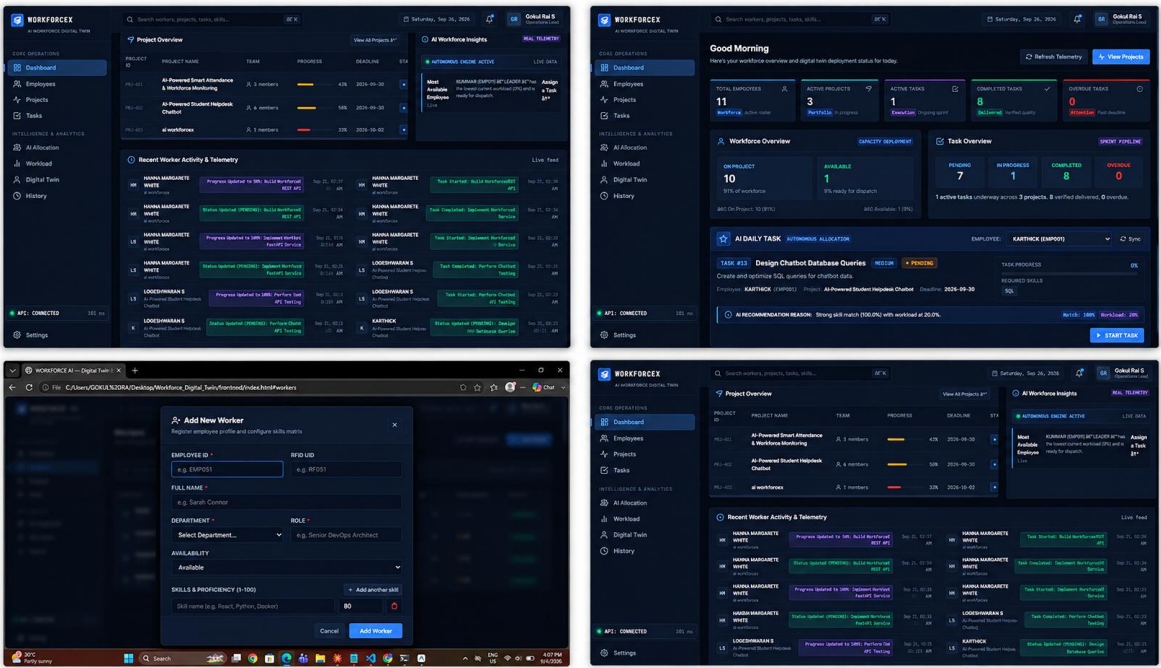


Fig 6.2 software (frontend ~AI integration)

Discussion

The results indicate that WORKFORCEX can provide a common platform for connecting **physical employee identification with digital workforce information**. Instead of maintaining employee, skill, workload, and project information separately, the system brings these elements together and makes them available through a single dashboard.

A key outcome is the ability to connect **employee skills with project requirements and workload conditions**. This can support more systematic task allocation and provide managers with better visibility into workforce utilization.

However, the current implementation is a **prototype**, and the AI allocation mechanism should be further validated using larger and more diverse employee, project, and task datasets. Future versions can incorporate real-time workload updates, more advanced optimization algorithms, predictive workforce analytics, and integration with existing enterprise systems.

Overall, the prototype demonstrates the **technical feasibility of WORKFORCEX as an AI-driven workforce digital twin**, while providing a foundation for further development toward real-world workforce planning and skill deployment applications.

CHAPTER 7

CONCLUSION AND FUTURE WORK

CONCLUSION

The **WORKFORCEX – AI Workforce Digital Twin & Skill Deployment System** provides an integrated solution for efficient workforce management by combining **RFID-based employee identification, real-time workforce monitoring, skill-based task allocation, workload analysis, and AI-assisted employee deployment**. The developed prototype successfully demonstrates the connection between the physical identification system and the software-based digital twin platform.

The system enables organizations to maintain employee profiles, monitor their availability and workload, manage projects and tasks, and identify suitable employees based on their skills and current workload. The **ESP32 and RFID module** provide a practical interface for employee identification, while the **FastAPI backend and dashboard** enable centralized workforce data management and visualization.

The prototype demonstrates that digital workforce representation combined with intelligent skill matching can improve visibility into workforce availability and support more structured task allocation. The developed system can serve as a foundation for a scalable workforce management platform that can be further enhanced using advanced AI and real-time organizational data.

FUTURE WORK

- **Advanced AI-based allocation:** Implement machine-learning models that continuously learn from employee performance, task completion time, workload, and skill development.
- **Real-time workforce analytics:** Integrate live attendance, task progress, and workload data to provide more accurate workforce insights.
- **Predictive workforce planning:** Predict future skill requirements and identify potential workforce shortages before project execution.
- **Automated skill-gap analysis:** Compare employee skills with project requirements and recommend suitable training or upskilling programs.

- **Integration with organizational systems:** Connect WORKFORCEEX with existing HRMS, attendance, payroll, and project-management platforms.
- **Mobile and cloud deployment:** Develop mobile applications and cloud-based infrastructure for remote access and organization-wide deployment.
- **Scalability:** Expand the system to support large organizations with multiple departments, projects, locations, and thousands of employees.
- **Enhanced security:** Introduce encrypted communication, role-based access control, secure authentication, and improved protection of employee data.
- **IoT expansion:** Integrate additional IoT sensors and devices for real-time workplace and workforce monitoring.
- **Performance prediction:** Use historical task and performance data to identify trends and support data-driven workforce planning.

6.1 Overall System Operation

The complete system begins with employee identification through an RFID card. Each registered employee is assigned an RFID card containing a unique identification number (UID). When the employee places the card near the RFID reader, the reader captures the UID and transfers it to the STM32F401RE. The STM32 processes the received information and compares it with the registered employee records. If the UID corresponds to a valid employee, the employee is authenticated and the attendance process is initiated.

After successful authentication, the system records the employee's attendance along with the corresponding date and time. The TFT display provides immediate information about the employee and the current attendance status. The LED and buzzer provide additional visual and audible confirmation, allowing the employee to know whether the RFID card has been accepted or rejected. This makes the system more interactive and reduces uncertainty during the attendance process.

The processed attendance information is then transferred from the STM32 to the ESP32. The ESP32 establishes Wi-Fi connectivity and acts as the communication interface between the embedded hardware and the software platform. The information received by the ESP32 can be transmitted to the backend or database, where it is stored and made available to the workforce dashboard.

6.2 Employee Identification and Attendance Result

The RFID-based identification mechanism provides a fast and contactless method of identifying employees. Unlike manual attendance registers, the proposed system does not require the employee or administrator to manually enter identification information. The unique RFID UID acts as the identification key for retrieving the corresponding employee record.

The attendance mechanism also improves data consistency because the employee information is obtained directly from the registered database. A valid card results in successful employee

identification and attendance recording, whereas an unregistered card can be treated as an invalid access attempt. Therefore, the system provides a basic authentication mechanism before attendance information is entered into the database.

The automated recording of attendance also provides a useful foundation for workforce availability monitoring. Once an employee has been successfully identified, the system can update the employee's current status on the dashboard. This allows administrators to distinguish between employees who are available and those who have not yet registered their presence.

6.3 Embedded Hardware Performance

The **STM32F401RE** serves as the primary controller responsible for coordinating the hardware components. It handles communication with the RFID reader and controls the output devices such as the TFT display, LED and buzzer. The use of a dedicated embedded controller allows time-critical operations such as RFID detection and local feedback to be performed without depending completely on the Internet connection.

The **ESP32** complements the STM32 by providing wireless communication capabilities. Separating local hardware control from network communication provides a practical architecture in which the STM32 can continue handling the physical interface while the ESP32 manages data communication with the software platform.

The TFT display improves the usability of the prototype by presenting information directly at the point of RFID interaction. The LED and buzzer provide simple status indicators, which are particularly useful for communicating successful authentication, invalid cards, or other system conditions.

6.4 Real-Time Data Communication

One of the important outcomes of the proposed system is the integration of the physical attendance unit with the digital workforce dashboard. Once an RFID transaction is processed, the corresponding information can be transferred through the ESP32 over Wi-Fi to the backend and database.

This communication eliminates the requirement for administrators to collect attendance information manually from the hardware unit. Instead, attendance information can be made available centrally for monitoring and further processing. The architecture also provides the possibility of extending the system to multiple attendance terminals in the future, where information from different locations can be transmitted to a common database.

The effectiveness of the complete system, however, depends on the availability and stability of the Wi-Fi network and backend services. A temporary network failure may prevent immediate synchronization with the dashboard even though local RFID identification can still be performed. This highlights the importance of maintaining local processing and providing suitable mechanisms for storing or retransmitting data when connectivity is restored.

6.5 Workforce Availability Monitoring

The proposed system extends beyond conventional attendance recording by using attendance information to determine workforce availability. The dashboard can classify employees according to their current status, such as available, unavailable, or assigned to a task.

This provides administrators with a real-time overview of the workforce instead of requiring them to manually check attendance registers or contact individual employees. Such information becomes particularly useful when a new task needs to be assigned quickly and the administrator needs to know which employees are currently available.

The combination of attendance and availability information therefore creates a link between **employee presence and workforce utilization**.

6.6 Skill Matching and Employee–Task Compatibility

A significant feature of the proposed system is the integration of skill information with workforce allocation. Each employee profile can contain information such as their department, skills, experience and availability. Similarly, each task can contain information regarding the skills required to complete it.

When a task is entered into the system, the Skill Matching and Task Assignment Module compares the task requirements with the skills possessed by available employees. A

compatibility score can then be calculated based on the degree of skill matching. Employees with higher compatibility scores can be ranked above employees with lower scores.

Fig 6.1 dashboard

For example, if a task requires **Python programming, database management and IoT knowledge**, the system can compare these requirements against the skill profiles of available employees. An employee possessing all three required skills would obtain a higher compatibility score than an employee possessing only one or two of them.

This approach provides a more systematic method of employee selection. Instead of relying entirely on manual judgment, the administrator receives a data-based recommendation. The system can also incorporate additional factors such as employee availability, experience and workload to improve the quality of the recommendation.

6.7 Dashboard Result

The workforce dashboard acts as the central monitoring interface of the proposed system. It provides administrators with a consolidated view of information collected from the embedded hardware and stored in the database.

The dashboard can include separate sections for:

- **Live Attendance** – displays employees who have registered their attendance.
- **Employee Management** – displays employee ID, name, department, skills and experience.
- **Workforce Availability** – identifies employees who are currently available.
- **Task Management** – displays active tasks and their required skills.
- **Skill Matching** – compares employee skills with task requirements.
- **Compatibility Result** – ranks employees according to their suitability for a task.
- **Task Assignment** – allows the administrator to assign the selected employee.
- **Attendance History** – maintains previous attendance records for reference.
- **Workforce Statistics** – presents useful information through tables, counts or graphs.

The dashboard therefore converts raw RFID attendance information into meaningful workforce information that can support administrative decision-making.

6.8 Integration of Hardware and Software

The overall integration can be represented by the following data flow:

RFID Card → RFID Reader → STM32F401RE → ESP32 → Wi-Fi → Backend/Database → Workforce Dashboard

The information flow begins at the RFID card and ends at the dashboard, while the local output is simultaneously provided through:

STM32 → TFT Display + LED + Buzzer

This architecture divides the responsibilities between the hardware and software components. The STM32 performs local control and peripheral management, while the ESP32 handles wireless connectivity. The backend and database manage data storage and processing, while the dashboard provides visualization and decision-support functionality.

This modular architecture also makes the system easier to expand. Additional RFID readers, sensors, employee profiles, tasks or software features can be incorporated without completely redesigning the system.

6.9 Limitations Observed

Although the proposed system provides an integrated approach, certain limitations remain. RFID-based identification depends on the physical RFID card being available to the employee, and card sharing could potentially affect the reliability of attendance records. Wireless communication also depends on network availability. Furthermore, the accuracy of the skill-matching result depends on the quality and completeness of the employee skill database.

The initial compatibility mechanism is based on predefined employee skills and task requirements. Therefore, it does not automatically determine whether an employee has actually developed a particular skill. Similarly, factors such as workload, previous task performance, learning ability and task complexity may require additional parameters for more advanced workforce allocation.

These limitations provide opportunities for future improvements without changing the basic architecture of the proposed system.

6.11 Overall Outcome

The proposed system successfully establishes a connection between **physical employee identification and digital workforce management**. The combination of RFID, STM32, ESP32 and the software dashboard provides a practical platform for automating attendance and monitoring workforce availability. The addition of skill matching and compatibility-based task allocation further increases the usefulness of the system by assisting administrators in selecting suitable personnel.

Overall, the project demonstrates that an IoT-enabled embedded system can be extended beyond conventional attendance applications to provide a **real-time workforce management and decision-support platform**. The modular design allows future integration of advanced features such as biometric authentication, AI-based skill prediction, workload balancing, performance analysis, predictive workforce planning and cloud-based multi-location management.

Discussion

The comparison between the existing and proposed workforce management systems demonstrates the advantages of integrating RFID, embedded controllers, IoT communication, and intelligent task allocation. The existing system primarily depends on manual attendance recording and conventional methods of employee management. Such methods require considerable human intervention and may result in delays, data-entry errors, and difficulty in obtaining the current status of employees. In contrast, the proposed system automates employee identification and attendance recording using RFID technology, thereby reducing manual effort and improving the speed of the attendance process.

The proposed system also provides **real-time workforce monitoring**, which is a major improvement over the existing approach. Attendance information collected through the RFID reader is processed by the STM32F401RE and communicated through the ESP32 to the software platform. This allows administrators to obtain updated information about employee presence and availability through the dashboard without depending on manual records. Consequently, the proposed system provides better visibility of the available workforce and supports faster administrative decision-making.

Another important improvement is the introduction of **skill-based employee-task matching**. In the existing system, task allocation may depend largely on manual decisions and the administrator's knowledge of employee capabilities. The proposed system maintains employee skill information and compares it with the requirements of a particular task. A compatibility score can then be generated to rank suitable employees. This provides a systematic approach to task assignment and helps ensure that employees with appropriate skills are considered for specific tasks.

The proposed system also improves **data accessibility and organization**. Instead of maintaining separate physical attendance records, employee details and task information can be stored in a centralized database. The dashboard provides a single interface through which authorized users can access attendance records, employee information, workforce availability, task requirements and compatibility results. Historical information can also be retrieved when required for analysis and decision-making.

The comparison therefore indicates that the proposed system provides improvements in **automation, response time, workforce visibility, task-assignment support and centralized data management**. However, the numerical values shown in the comparison graph should be considered **illustrative unless they are obtained from actual experimental testing**. For the final project report, the values should ideally be replaced with measurements collected from the developed prototype.

Overall, the discussion demonstrates that the proposed system is not limited to conventional attendance management. It extends attendance information into a broader **workforce intelligence platform**, where employee presence and skill information can be utilized to support efficient task allocation. This integration of hardware-based identification, IoT connectivity and software-based workforce analysis represents the primary contribution of the proposed system.

CHAPTER 7

CONCLUSION AND FUTURE WORK

Conclusion

The proposed **Smart RFID-Based Management System** is designed to provide an efficient and reliable method for identification, monitoring, and management using RFID technology and embedded systems. The system uses an RFID module to detect and identify authorized RFID cards or tags and processes the obtained information through the microcontroller. The **STM32F401RE** acts as the main controller for handling the system operations, while the **ESP32** provides communication and connectivity features. The system displays the relevant information on the TFT display and provides visual and audio indications through the LED and buzzer.

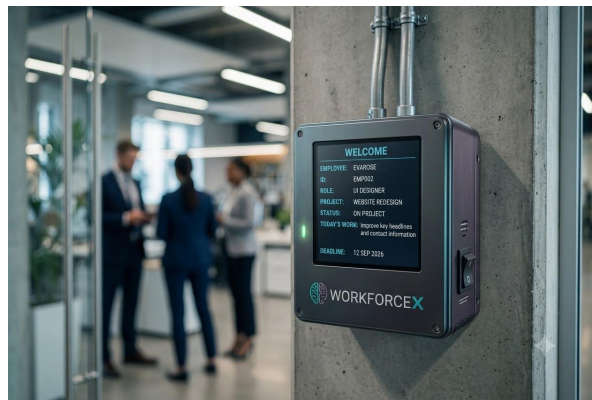


Fig7.1 workforcex in real time.

The developed system offers real-time identification and monitoring with a simple and user-friendly interface. It reduces manual effort, minimizes errors associated with conventional identification methods, and improves the speed and efficiency of the overall process. The integration of RFID, microcontrollers, display, indicators, and wireless communication makes the system suitable for practical applications where secure and automated identification is required. Overall, the proposed system provides a compact, reliable, and cost-effective solution for smart management and identification applications.

Future Work

In the future, the **Smart RFID-Based Management System** can be further improved by integrating advanced technologies to enhance its functionality, security, and reliability. The

system can be connected to a cloud-based database for storing and managing RFID records remotely. A web or mobile application can also be developed to allow authorized users to monitor system activities and access records in real time.

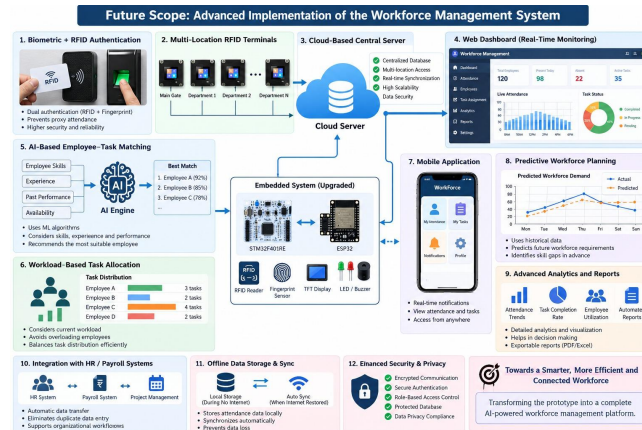


Fig 7.2 future scope advanced implementation

In addition, advanced security features such as encrypted RFID communication, user authentication, and multiple-level access control can be incorporated to improve system security. The system can also be enhanced by adding additional sensors and communication technologies such as Wi-Fi and Bluetooth for wider applications. Future developments may also include automated report generation, data analytics, remote monitoring, and improved display interfaces. These enhancements can make the proposed system more scalable, secure, and suitable for real-world smart management applications.

Sources

To pick up a draggable item, press the space bar. While dragging, use the arrow keys to move the item. Press space again to drop the item in its new position, or press escape to cancel.

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